Interlayer origin of superconductivity in $Na_xCoO_2 \cdot yH_2O$

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At the compositions from which its hydrate is prepared, anhydrous Na_xCoO_2 is a paramagnetic metal at all measured temperatures, yet intercalating a bilayer of water between its CoO_2 sheets produces a superconductor with $T_c \approx 4.5$ K, and two decades of theoretical work, largely confined to the CoO_2 sheet, have not yielded an accepted mechanism. Using density-functional calculations motivated by our earlier prediction of a topological two-dimensional electron gas at cobaltate surfaces, we show that hydration reproduces that surface electronic structure in the bulk. Expanding the interlayer gallery beyond a critical spacing $c^* \approx 6.4$ Å has two simultaneous consequences: a two-dimensional electron gas condenses on the sodium plane, and the sodium site potential bifurcates from a single stiff well into a shallow double well. The gas supplies the carriers, the quantized anharmonic mode supplies the pairing boson, and an Allen–Dynes treatment of the parity-allowed quadratic coupling yields a superconducting dome (peak $T_c \approx 14$ K, range 11–23 K) bounded by a carrier-depleted wall below c^* and a polaronic wall above it, correctly excluding both the anhydrous compound and the monolayer hydrate. The superconducting bilayer hydrate, at $c = 9.9$ Å, lies beyond this static window: there the water cage rebinds the sodium, adiabatic relaxation deepens rather than flattens the well, and the required softening must therefore be statistical and dynamical rather than adiabatic, a conclusion consistent with the null H/D and $^{16}O/^{18}O$ isotope shifts. The pairing thus resides in the hydrated gallery rather than in the CoO_2 sheets.

The discovery of superconductivity in the bilayer hydrate $Na_xCoO_2 \cdot yH_2O$ ($T_c \approx 4.5$ K) [1] posed a question that remains open after two decades. The anhydrous compound at the same composition ($x \approx 0.3$) is a paramagnetic metal down to the lowest measured temperatures, and superconductivity appears only when a bilayer of water is intercalated between the CoO_2 sheets; no accepted mechanism explains why the water is essential [2].

First-principles studies addressed this question early and reached a consistently negative verdict: with the water held static, hydration leaves the CoO_2-derived electronic structure essentially unchanged beyond the expected effects of lattice expansion [3–5]. These calculations share two assumptions: that the relevant physics resides within the CoO_2 sheet, and that the water is inert.

Our starting point lies outside the sheet. In earlier work on the layered cobaltates we showed that the topology of an obstructed atomic limit, seeded by an annihilating Dirac cone [6, 7], requires the half-open gallery at the surface to host a spin-split two-dimensional electron gas residing entirely outside the CoO_2 sheet, its density centred on the surface alkali ions [8]. Here we show that hydration opens the same gallery in the bulk. Beyond a critical interlayer spacing, charge held by the oxygen planes returns to the gallery as the predicted two-dimensional gas, and at the same spacing the sodium site potential, previously a single stiff well, bifurcates into a shallow double well. The former supplies carriers, the latter a soft anharmonic mode that mediates their pairing, and a superconducting dome occupies the window in which the gas is present and the sodium is soft but not yet self-trapped. In this picture the superconductivity of the hydrate resides not in the CoO_2 sheets but in the gallery between them, which is why it appears only upon hydration.

The transition. Consider a sodium ion midway between two CoO_2 sheets as the interlayer spacing is increased. For narrow galleries the midplane is the minimum of a single well; beyond a critical spacing it becomes a local maximum separating two equivalent off-centre minima, one toward each sheet, between which the ion oscillates. The electronic structure changes at the same spacing: charge held by the oxygen planes transfers into the gallery as a two-dimensional gas whose density envelops the sodium plane. We compute both effects with spin-polarized density-functional theory (PBE [9], projector-augmented waves [10], Quantum ESPRESSO [11]), fitting $E(\delta) = E_0 + \alpha\delta^2 + \beta\delta^4$ to the total energy of the off-centre sodium at each spacing c (methods and convergence, SM S1 [12]; well data and fits, SM S2). The sign of α identifies the regime: positive at $c = 5.5$ Å ($\alpha = +2.5$ eV/Å^2, a single stiff well), negative by $c = 6.9$ Å ($\alpha = -0.5$ eV/Å^2, a double well), with the crossing at $c^*_{\rm Na} \approx 6.40$ Å (full-coverage cell; the physical $x = 1/3$ cell shifts the crossing by only ≈ 0.2 Å, SM S2). The Fermi-level density of states $N(0)$ tracks the same crossing, increasing by a factor of 30 between 5.5 and 6.9 Å and saturating beyond it, and the Bader charge returned to the sodium grows monotonically from 0 to $0.58\,e$ over the same range. That charge corresponds to an areal density $n_{\rm 2D} \approx 8 \times 10^{14}$ cm^{-2}, whose implied Fermi radius $k_F = \sqrt{2\pi n_{\rm 2D}} \approx 0.7$ Å^{-1} matches the Fermi-surface barrel computed independently at 9.9 Å (SM S4, S5). These are two signatures of a single event. In the closed gallery the sodium electron resides in the

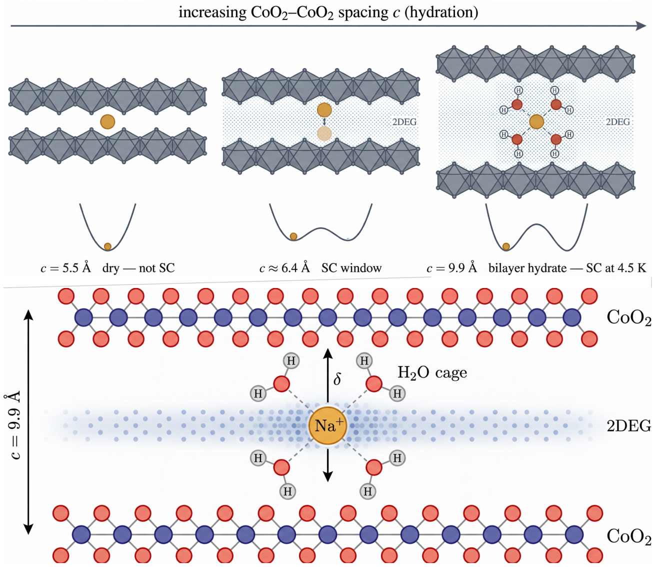

FIG. 1. Schematic of the mechanism as a function of the gallery spacing c (top row): in the anhydrous compound the sodium occupies a single stiff well and no interlayer gas is present, so there is no pairing; within the calculated window the well bifurcates at the same spacing at which the 2DEG condenses, and the even overtone of the quantized well supplies the pairing boson; at 9.9 Å, outside the static window, the water itself resoftens the sodium potential (Fig. 4). Bottom row: the bilayer hydrate in detail, showing the four-water cage, the sodium's c-axis displacement δ, and the interlayer 2DEG, whose density is centred on the sodium plane and envelops the ion (the band is 91% Na in character; see Fig. 3).

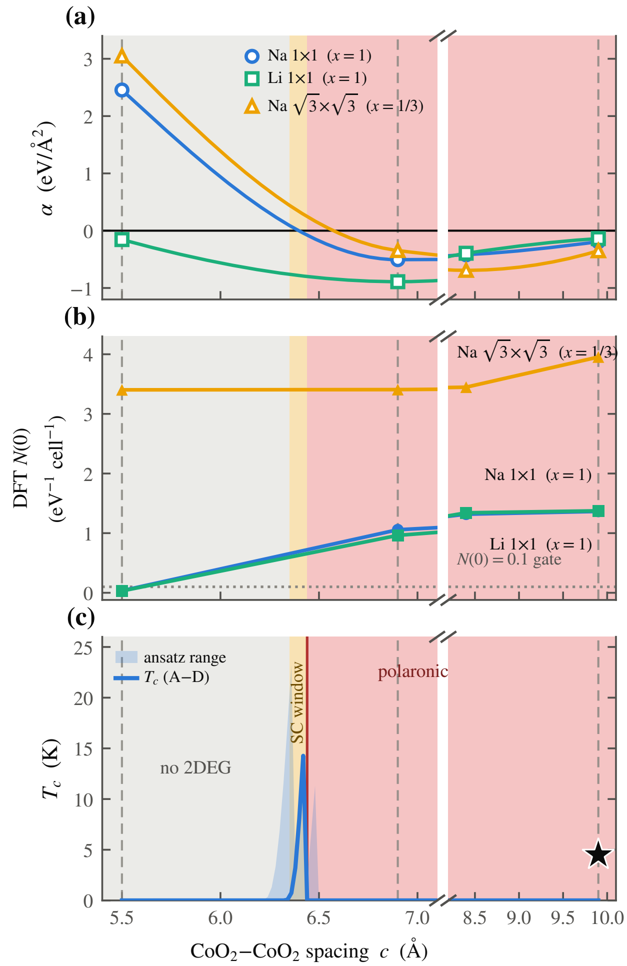

FIG. 2. Well bifurcation, carrier onset, and the superconducting dome as functions of the CoO_2–CoO_2 spacing c (axis broken to place all four measured spacings on one scale). (a) The Landau coefficient $\alpha(c)$ crosses zero at $c^*_{\mathrm{Na}} \approx 6.40$ Å. (b) The Fermi-level density of states $N(0)$ turns on at the same spacing. (c) The Allen–Dynes $T_c(c)$: a narrow dome (shaded band, ansatz range 11–23 K) between the carrier-depleted wall (left) and the $\lambda = 2$ polaronic wall (vertical line), beyond which pairing is suppressed. Dashed verticals mark the anchors at 5.5, 6.9, and 9.9 Å: the anhydrous parent and the monolayer hydrate fall on the walls and are correctly non-superconducting; the bilayer hydrate (star, $T_c = 4.5$ K) lies outside the static window, a discrepancy resolved in Fig. 4.

Na–O bond, and that bond is what holds the ion centred; opening the gallery returns the electron to the sodium, where it spreads into the interlayer gas, and the ion, no longer bound by the bond, becomes bistable. That the charge, not the spacing itself, is the agent is shown by two controls (SM S12): adding 0.1–0.3 e per formula unit to the closed (5.5 Å) gallery through a compensating background (lithium cell) fills the analogous interlayer band and softens the well at fixed lattice constants, and the well stiffness responds directly to the oxygen-height channel that carries the transfer, $d\alpha/dz_{\mathrm{O}} \approx -0.52$ eV/Å^2 per Å.

Figure 2 collects $\alpha(c)$, $N(0)$, and the superconducting dome constructed from them on a common axis with a broken scale spanning all four measured spacings. Inserting $\lambda(c)$, the boson energy, and $N(0)$ into the Allen–Dynes formula (coupling model below, $\mu^* = 0.10$; carrier gate and polaron cut, SM S1) yields a dome pinned just above c^*, with peak $T_c \approx 14$ K, that correctly excludes both non-superconducting anchors: the anhydrous compound falls on the carrier-depleted side and the monolayer hydrate beyond the polaronic wall. Two systematics bound these numbers: holding the oxygen planes at fixed height carries a well-depth uncertainty of about $\pm 50\%$, which changes neither the sign of α nor any verdict in Table I, and the peak T_c of 14 K (range 11–23 K, against the observed 4.5 K) fixes an order of magnitude rather than a precise value. Photoemission on the parent finds bands narrower than the Kohn–Sham ones by roughly a factor of two; that correction raises $N(0)$ and with it λ, so the existence of the dome does not rest on an inflated coupling. The direction of this and every other correlation correction is catalogued in SM S10.

The gallery band. The band structure exhibits the same crossing directly. Figure 3 tracks the Kohn–Sham spectral weight along Γ–M–K–Γ at three spacings, with the sodium character drawn as a blue fatband. At $c = 5.5$ Å the sodium band lies approximately 2 eV above E_F and carries no weight there; by $c = 9.9$ Å a band of 91% sodium character has descended through E_F; it is the interlayer gas predicted at the surface [6], now realized in the bulk gallery. The resulting Fermi surface (SM S4) is a single electron barrel around Γ with no e'_g pockets. The hydrate itself has never been measured, because it dehydrates under ultrahigh vacuum, so the only reference is the anhydrous parent, where photoemission finds a Co-derived a_{1g} barrel of nearly the same k_F and likewise no e'_g pockets [13]. The comparison carries two messages. The absence of the e'_g pockets, on which several in-sheet pairing proposals were built, is an experimental fact already at the parent level. And in the present picture hydration exchanges the orbital identity of the barrel, cobalt for 91% sodium, while nearly preserving its footprint, so the two Fermi surfaces are indistinguishable at the level of topology and size, one reason the gallery band could escape notice. The discriminating test is therefore orbital character: photoemission on freshly hydrated crystals should find the barrel rebuilt from sodium weight, together with a 20–30 meV self-energy feature from the confined sodium mode (the parent shows a kink onsetting near 26 meV [13]; its gallery is closed, so we do not identify the two).

The pairing window. The remaining ingredient is the coupling of this band to the sodium mode. The two bands sharing the gallery electron, alkali sp_z and CoO_2, form the SSH-type pair of our earlier surface model [6], split by the same Bader charge that fills the gallery band in Fig. 2(b). The sodium displacement δ is odd under reflection through the gallery midplane, so the linear electron–phonon coupling vanishes identically at the symmetric point (verified at several k_z). The leading coupling is quadratic: the displaced ion admixes the band of opposite parity, polarizing its own cloud against the motion, and the second-order sum over these virtual transitions gives $\chi \approx -6$ eV/Å^2 near the transition (SM S3). An electron therefore scatters off two quanta of the mode rather than one, and the boson is the even overtone $\hbar\Omega = E_2 - E_0$ of the quantized well (the transition to the first excited state is forbidden by the same reflection). The dimensionless coupling is

$$\lambda = \frac{2\,N(0)\,g^2}{\hbar\Omega}, \tag{1}$$

with $N(0)$ from the calculation, not fitted, and g the quadratic vertex (SM S3).

The window is bounded on both sides by elementary physics. Below c^*, $N(0) \to 0$ and pairing fails for lack of carriers. Well above it, or whenever the sodium localizes in a single well, the ion self-traps while $\hbar\Omega$ remains near 20 meV, so λ exceeds 2 and pairing is suppressed in the polaronic regime instead. Between the two walls the quantized well is soft, its effective zero-point frequency collapsing from 30 meV in the stiff single well to a fraction of a meV in the double well (SM S6), and the Allen–Dynes dome of Fig. 2(c) occupies the resulting window, 6.35–6.45 Å, with neither the monolayer nor the bilayer anchor inside it. The position of the bilayer hydrate is the central difficulty, which we address next.

The bilayer hydrate and the water. The bilayer hydrate is the only member of the family that superconducts, and it holds its CoO_2 layers 9.9 Å apart [14], well outside the window calculated above. Moreover, the bare sodium potential at 9.9 Å is not a soft double well at all: the energy decreases monotonically as the sodium approaches one CoO_2 sheet, an instability rather than a bounded anharmonic mode. The static calculation therefore places the only superconducting member of the family in the polaronic regime. This is the inert-water null result of Refs. [3, 5] reappearing at the level of the lattice dynamics, and it indicates that the water must be treated dynamically.

We constructed the bilayer solvation cage that neutron diffraction finds around the sodium [14, 15], four rigid H_2O per cell, and recomputed $E(\delta)$ against the same cell with the water removed (construction, SM S7). Figure 4(a) shows the result. Without water the potential remains unstable; with the rigid cage the sodium is rebound into a double well with a confined mode at $\hbar\omega_{\text{eff}} \approx 23$ meV, finite and positive where the bare mode was imaginary. Allowing the same four waters to relax at each sodium position does not flatten this well but deepens it, from 199 meV (frozen cage) to 380 meV (adiabatic), with a stiffer mode near 37 meV; a dispersion-corrected calculation reproduces the trend. Adiabatically relaxed water therefore binds the sodium more strongly, not less. The softening must instead be sought in the width of the cage-configurational ensemble: at fixed δ the hydrogen-bond network presents a spread of local minima at least 0.4 eV wide [Fig. 4(a), amber band], and the relevant time scales are well separated: the sodium mode has a period near 200 fs, while the network reorganizes on the picosecond scale measured by quasielastic neutron scattering [16]. Fast on the time scale of the cage, the sodium samples the softer instantaneous members of this ensemble rather than following the adiabatic surface to its deep minimum. The softening required by the mechanism is therefore statistical rather than adiabatic, and the decisive open test is a finite-temperature Born–Oppenheimer or path-integral molecular-dynamics sampling of the coupled sodium–water system (SM S7).

This picture makes a definite prediction for isotope substitution, and the prediction has already been tested. Replacing H_2O by D_2O shifts T_c by less than 0.2 K [17], and substituting ^{18}O for ^{16}O is likewise null [18]. A mechanism in which electrons pair through a water vibration is difficult to reconcile with these results; the present mechanism requires them, because the pairing boson is the

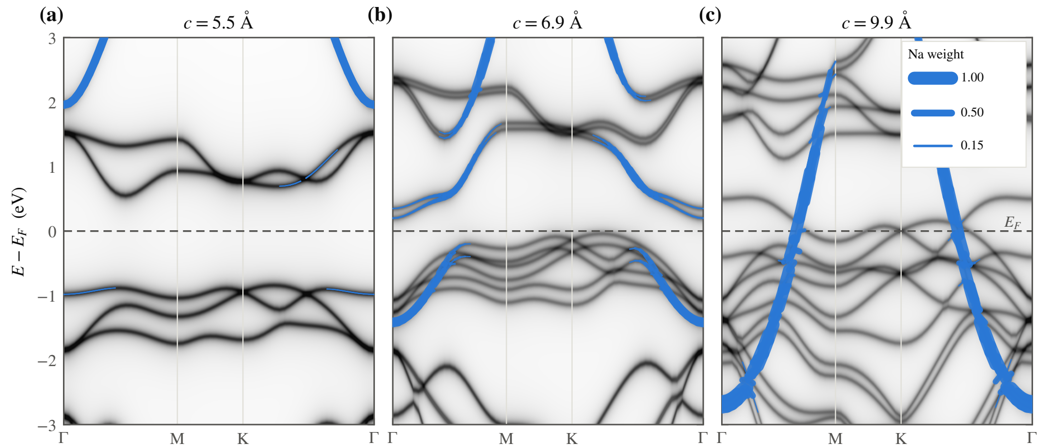

FIG. 3. Emergence of the gallery band. Kohn–Sham spectral weight along Γ–M–K–Γ at (a) 5.5, (b) 6.9, and (c) 9.9 Å, referenced to E_F (dashed); grayscale gives the total weight, the blue fatband the sodium-projected weight. At 5.5 Å the cell is a band insulator with the sodium band 2 eV above E_F; by 9.9 Å a band of 91% sodium character crosses E_F, the interlayer two-dimensional gas appearing as a single band and realizing in the bulk the surface prediction of Ref. [6]. Fermi-surface topology and the ARPES comparison are given in SM Sec. S4.

sodium mode and the water mass enters only through the anharmonicity of the confining potential. No corresponding test exists for the sodium itself, ^{23}Na being its only stable isotope; the soft ∼20 meV Na c-axis mode must instead be sought spectroscopically, by inelastic neutron scattering or low-frequency Raman scattering on freshly hydrated crystals, where it should appear as a dynamical mode rather than a static site splitting.

Consequences. Lithium cobaltate, the same structure with a lighter gallery ion, has never been observed to superconduct, and the present picture requires this. Its well crosses into the double-well regime only at the smallest spacing we sampled, where it is only 4 meV deep while the lithium zero-point energy is 8.8 meV, so the ion remains delocalized across both minima, a quantum paraelectric with no soft anharmonic mode available for pairing. A heavier gallery ion should shift the threshold outward: for potassium the crossing moves to $c^*_{\mathrm{K}} \approx 6.75$ Å, ordering Li $<$ Na $<$ K with ionic size, a prediction for hydrated K_xCoO_2. Immediately above the carrier onset the gallery band is narrow enough to be Stoner-polarized, producing a thin magnetic strip adjacent to the dome (SM S8, where the computed regimes are assembled into a phase diagram); this accounts for the observed proximity of magnetism to the superconducting phase and suggests that the conflicting NMR results, triplet pairing inferred from powder samples and singlet from single crystals, reflect where each sample lies relative to the Stoner boundary rather than a genuine contradiction.

The same charge return leaves a valence fingerprint: a site-specific probe (Co L-edge, NQR) and a charge-counting probe (titration) should disagree by $\Delta v \approx 0.06$ per cobalt only once the gallery is open, distinct from the larger valence offsets produced by an oxonium reservoir [19]. Electrostatic gating of a $LiCoO_2$ film should fill the gallery to $n_{2\mathrm{D}} \approx 4 \times 10^{14}$ cm^{-2} while the light ion remains centred, a test of the carriers in isolation from the pairing mode, the experimental counterpart of the gating control above (SM S12). Na_2CoSe_2O [20] superconducts at 6.3 K through its Co–Se network; we find its gallery band unoccupied, so it demonstrates the distinction the picture requires: a cobaltate can superconduct without the gallery, but then not through it (SM S12). The pairing state favored here is a time-reversal-symmetric singlet $s_\pm$ spanning the gallery gas and the a_{1g} pocket, and the measurements that eliminated the principal in-sheet proposals leave it intact: the e'_g pockets on which the triplet and f-wave states were built [21, 22] are absent in photoemission [13], the single-crystal Knight shift excludes the triplet channel [23], the μSR null excludes the chiral states [24, 25], and the isotope nulls exclude any water-vibration glue [17, 18]. The gallery mechanism is consistent with all four (SM S9).

Outlook. Five measurements, listed roughly in order of increasing difficulty, would test the mechanism directly: inelastic neutron scattering or low-frequency Raman scattering for the soft, dynamical ∼20 meV sodium mode in the fresh hydrate; character-resolved photoemission on hydrated crystals for the sodium weight of the barrel and its 20–30 meV self-energy feature; x-ray absorption or NQR for the valence split; electrostatic gating of $LiCoO_2$ films for gallery carriers with the ion left centred; and hydrated K_xCoO_2 at the predicted 6.75 Å threshold. On the theoretical side, the decisive calculation is a finite-temperature ab initio or path-integral molecular-dynamics sampling of the coupled sodium–water system. If the picture survives these tests, it implies a design rule broader than this compound: expand-

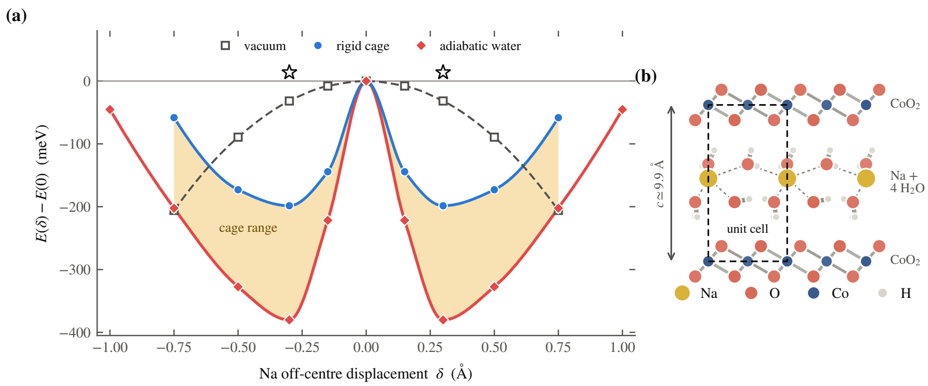

FIG. 4. Effect of the water cage on the sodium potential. (a) $E(\delta)$ of the off-centre sodium at $c = 9.9$ Å: with the water removed (open squares) the sodium is unstable toward chemisorption on the CoO_2 sheet; a rigid water cage (blue) rebinds it into a double well, $\hbar\omega_{\mathrm{eff}} \approx 23$ meV; allowing the cage to relax (adiabatic, red) deepens the well further, to 380 meV, with a stiffer 37 meV mode. The amber band is the $\gtrsim 0.4$ eV spread of cage-configurational minima at fixed δ; the open stars (one computed point, mirrored as the even curves are) mark a partially relaxed configuration caught in one shallower member of that spread. The sodium (∼200 fs) samples this ensemble rather than the slow (∼ps) adiabatic surface, so the softening is statistical (text; SM S7). (b) The hydrate cell along c: CoO_2 sheets, the sodium plane, and the two-layer water cage between them.

TABLE I. The mechanism against the experimental record. "Explained" follows from the static calculation; "passed" is a pre-existing null result the mechanism requires; "matched" is a computed quantity against a measured one. The asterisk marks the one contingent entry: the static geometry at 9.9 Å is polaronic, and the required softening is the dynamical water effect of Fig. 4 (SM S7). Line-by-line derivations and the comparison with competing proposals are given in SM S9.

Observation	**Verdict**
Anhydrous (5.5 Å) not SC [26]	explained
Monolayer hydrate (6.9 Å) not SC [26]	explained
Bilayer hydrate SC at 4.5 K [1]	explained*
T_c a few K, not tens of K [1]	explained
No superconducting Li analogue	explained
Magnetic phase borders the dome [23]	explained
Singlet–triplet NMR conflict [23]	explained
H/D and $^{16}O/^{18}O$ nulls [17, 18]	passed
Γ-barrel footprint, no e_g' pockets [13]	matched

ing a layered gallery beyond the spacing at which its donor ion becomes bistable simultaneously releases the ion's electrons into an interlayer gas and softens the ion into the pairing boson, supplying both carriers and coupling from a single structural parameter.

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